Infinite NPA depth, finite semidefinite lift

An exact quantum region in the simplest Bell experiment

The quantum region of the CHSH value and the sum of two marginal averages requires infinite Navascués–Pironio–Acín depth, yet admits an exact finite semidefinite lift. We construct this lift by showing that every supporting bound is attained with equal measurement overlaps: one angle determines the entire boundary. Finite NPA certificates must remain valid for two independently varying angles, where the obstruction to finite depth survives. The same region has the smallest semidefinite extension degree possible for a nonpolyhedral set, separating infinite hierarchy depth from minimal semidefinite complexity.

Alice and Bob share a quantum state and choose between observables A_x, B_y with outcomes ± 1, where $x, y \in \{0, 1\}$. Write $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$ and $E_{xy} = \langle A_x B_y \rangle$. Retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \tag{1}$$

Let $\mathcal{W}$ contain all pairs (M, S) attainable with quantum states and local measurements. The marginal averages need not be equal. For its level-L NPA outer approximation $\mathcal{W}_L$, the preceding Letter [1] and the construction below give

$$\boxed{\begin{array}{c}\mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \\ \operatorname{sxdeg}(\mathcal{W}) = 2.\end{array}} \tag{2}$$

A semidefinite lift describes membership by positive semidefinite (PSD) matrices whose entries are affine in (M, S) and auxiliary variables. We construct one with matrix sizes five and four. The extension degree is the smallest possible largest matrix size; its value two follows from a theorem on planar convex sets.

Why one angle suffices. To determine $\mathcal{W}$, consider an arbitrary supporting inequality $hS + \alpha M \le \lambda$, with real h, α. After realizing the measurements projectively on a larger space, Jordan's lemma reduces this question to whether the two-qubit matrices $\lambda I - H$ are PSD. Here

$$H = h\,\mathrm{CHSH} + \alpha(ZI + IZ),$$
$$A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X.$$

Here X, Z are Pauli matrices, $A_0 = B_0 = Z$, and $\mathrm{CHSH} = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1$. Products of local operators mean tensor products. The numbers $c, d \in [-1, 1]$ are the cosines of the angles between each party's two measurement directions. Deterministic strategies attain $\beta = 2|h| + 2|\alpha|$, so $\lambda \ge \beta$ is necessary. Above this bound, the first three elementary symmetric functions of the eigenvalues of $\lambda I - H$ are nonnegative. Its determinant therefore decides positivity: if it too is nonnegative, the characteristic polynomial of $H - \lambda I$ has nonnegative coefficients and hence no positive root.

Put $x = (1 - c)/2$, $y = (1 - d)/2$, and write $D(x, y) = \det(\lambda I - H)$. The determinant takes the form

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \tag{3}$$

where F depends on the product alone. In the nonclassical range $|\alpha| \le |h|$, the coefficient of $x + y$ is nonnegative. At fixed product, $x + y \ge 2\sqrt{xy}$, with equality at $x = y = \sqrt{xy}$. Thus equal overlaps minimize the determinant; positivity there implies positivity for every pair. For $|\alpha| \ge |h|$, the maximum is the local bound β: average the Alice and Bob tilted-CHSH bounds at their local threshold, then use $\pm(A_0 + B_0) \le 2I$ for the remaining marginal coefficient. Equal overlaps attain this value too, for every real h, α.

At $c = d$, use the Bell states $\phi_\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $\psi_\pm = (|01\rangle \pm |10\rangle)/\sqrt{2}$. The singlet ψ_- decouples. In the ordered triplet basis (ϕ_+, ϕ_-, ψ_+), set $c = (1 - t^2)/(1 + t^2)$ and multiply $\lambda I - H$ by $(1 + t^2)^2$. With

$$p_0 = (1+t^2)^2, \qquad p_1 = 1-t^4, \qquad p_2 = 1+4t^2-t^4,$$

the result is the quartic matrix

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \tag{4}$$

Its value at zero and leading coefficient force $\lambda \ge \beta$, so the singlet bound holds too. Continuity includes $c = -1$, while $K(-t) = RK(t)R$, $R = \mathrm{diag}(1, 1, -1)$, includes negative t. We have therefore proved

$$\max_{\mathcal{W}}(hS + \alpha M) \le \lambda \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \tag{5}$$

The entries are affine—constant plus linear—in λ, α, h. Thus one fixed matrix describes every supporting inequality.

From one variable to one finite lift. A real matrix polynomial in one variable is PSD everywhere exactly when it equals $\sum_j Q_j(t)^T Q_j(t)$ for polynomial matrices Q_j, including singular cases [2]. Quartic diagonal entries force each Q_j to have degree at most

two. Thus, for $\xi \in \mathbb{R}^3$, $\xi^T K(t)\xi$ is a sum of squares of linear combinations of $t^k\xi_i$, $0 \le k \le 2$. Averaging under $(t,\xi) \mapsto (-t, R\xi)$ cancels terms that change sign and leaves two groups of monomials:

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T,$$
$$Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Consequently Eq. (5) is equivalent to the polynomial identity

$$\xi^T K(t)\xi = Z_+^T G Z_+ + Z_-^T J Z_-, \qquad G \succeq 0, \quad J \succeq 0, \tag{6}$$

The PSD coefficient matrices G, J, of sizes five and four, are called Gram matrices. Matching coefficients gives sixteen affine equations.

Let $\mathcal{L}_z$ be a linear functional on the sixteen-dimensional span of the entries of $Z_+Z_+^T, Z_-Z_-^T$, specified by sixteen real values z. Apply it entrywise to define

$$U(z) = \mathcal{L}_z(Z_+Z_+^T), \quad V(z) = \mathcal{L}_z(Z_-Z_-^T),$$

and define ℓ, m, s by $\mathcal{L}_z(\xi^T K\xi) = \lambda\ell(z) - \alpha m(z) - hs(z)$. Then the membership test is

$$\mathcal{W} = \{(m(z), s(z)) : \ell(z) = 1,\ U(z), V(z) \succeq 0\}. \tag{7}$$

The complete proofs list every matrix entry.

There is a direct physical proof of the reverse inclusion. For a real unit triplet state ξ at angle t, choose

$$U = \frac{Z_+Z_+^T}{(1+t^2)^2}, \qquad V = \frac{Z_-Z_-^T}{(1+t^2)^2}.$$

These matrices come from $\mathcal{L}_z(P) = P(t,\xi)/(1+t^2)^2$, with $\ell(z) = 1$ and coordinates equal to the state's M, S. Their limits include $t = \infty$. The triplet states attain every quantum maximum of $hS + \alpha M$; the omitted singlet points $(0, -2c)$ are mixtures of triplet endpoints $(0, \pm 2)$. Their convex hull is therefore $\mathcal{W}$. Mixing witnesses admits every point. Conversely, pairing any feasible matrices with every Gram certificate gives $\lambda - hS - \alpha M \ge 0$, so no point outside $\mathcal{W}$ is admitted.

The set of pairs obtained from triplet states is compact and connected. A point in its planar convex hull lies on a chord joining two image points: otherwise its connected radial image would lie in an open semicircle, giving a separating line. Thus every point of $\mathcal{W}$ needs at most two real equal-overlap strategies, with both membership matrices of rank at most two.

Why the NPA obstruction remains. Equal overlaps determine support values; a finite NPA certificate must represent the bound for independently chosen measurement angles. Write $f_L(\alpha)$ for the NPA maximum and $f(\alpha)$ for the quantum maximum of $S + \alpha M$. As $\alpha \uparrow 1$, a bounded ratio $[f_L - f]/[f - (2+2|\alpha|)]$ would force a shifted Motzkin polynomial to be a sum of squares [1]. The numerator is the NPA error; the denominator is the quantum violation above the local bound. The complete earlier argument is retained in the companion.

Minimal block order and thirty-two exact projections. For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the exact quantum bound has the expansion [1]

$$f(1-T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0. \tag{8}$$

For a polygon, this maximum would be affine sufficiently close to $\alpha = 1$, contrary to Eq. (8). Thus $\mathcal{W}$ is nonpolyhedral. Its finite lift makes it semialgebraic: it is a finite union of sets defined by polynomial equalities and inequalities. Since it is also closed and convex, Scheiderer's planar theorem gives $\mathrm{sxdeg}(\mathcal{W}) \le 2$ [3]. Order-one lifts are polyhedral, hence $\mathrm{sxdeg}(\mathcal{W}) = 2$. The displayed lift has sizes $5, 4$; the theorem guarantees a lift using matrices of size at most two.

Physical input and output relabellings give the same quantum region for

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij}\right), \qquad \prod_{ij} \chi_{ij} = -1.$$

Here $\chi_{ij} \in \{\pm 1\}$. The eight CHSH sign patterns and four selected input pairs give 32 distinct planes. Intersecting their fixed lifts with any NPA relaxation makes every support value on these planes exact, including all the directions responsible for finite-level nontermination.

References

[1] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, manuscript (2026). Its complete nontermination proof is given in the companion, Part II.

[2] C. Hanselka and R. Sinn, *Math. Z.* **292**, 83–101 (2019), Corollary 2.8; arXiv:1707.08261.

[3] C. Scheiderer, *SIAM J. Appl. Algebra Geom.* **5**, 114–139 (2021), Theorem 0.1; arXiv:2004.04196.